Scott W. Linderman

Contact Information

Wu Tsai Neurosciences Institute Room M252G
290 Jane Stanford Way
Stanford University
Stanford, CA 94305-4020

email: scott.linderman@stanford.edu
web: https://lindermanlab.github.io/

Professional Experience

Stanford University
Associate Professor, Department of Statistics — 2026-
Assistant Professor, Department of Statistics — 2019-26
Faculty Scholar, Wu Tsai Neurosciences Institute
Faculty Affiliate, Stanford Bio-X and Stanford AI Lab
Co-Director, Stanford Center for Neural Data Science

Engram — 2025-
Co-Founder and Chief Scientist
www.engram.com

Allen Institute — 2025-
Affiliate, Studio for Data Driven Discovery

Google — 2022
Visiting Faculty Researcher, Google Brain
Host: Kevin Murphy

Columbia University — 2016-19
Postdoctoral Fellow, Department of Statistics
Advisors: Liam Paninski and David Blei

Microsoft — 2008-11
Software Development Engineer in Test

Education

Harvard University — 2011-16
Ph.D., S.M., Computer Science
Advisors: Ryan Adams and Leslie Valiant

Cornell University — 2004-08
B.S., Electrical and Computer Engineering
Magna cum Laude with Honors in Engineering

Honors and Awards

- NSF CAREER Award — 2025
- McKnight Scholar Award — 2023
- Sloan Research Fellowship — 2022
- Allen Institute Next Generation Leaders Council — 2019
- Best Paper, Artificial Intelligence and Statistics (AISTATS) — 2017
- Leonard J. Savage Award, International Society for Bayesian Analysis — 2017
- Simons Collaboration on the Global Brain Postdoctoral Fellowship — 2016
- Siebel Scholarship — 2015
- National Defense Science and Engineering Graduate Fellowship — 2011

Publications

Note: all publications have authors listed in the order as published. Authorship convention in the field: lead senior author is listed last; lead junior author is listed first. (Asterisks denote joint lead or senior authorship).

Books and Theses

- **Scott W. Linderman**. *Machine Learning Methods for Neural Data Analysis*. 2024. *In preparation. Available at* `https://slinderman.github.io/ml4nd`

- **Scott W. Linderman**. *Bayesian methods for discovering structure in neural spike trains*. PhD thesis, Harvard University, 2016.
  Received the Savage Award from the International Society for Bayesian Analysis.

Preprints and Manuscripts Under Review

- E. Kelly Buchanan, Crystal Zhou, **Scott W. Linderman**, and Christopher Ré. Benchmarks as software: Terminal-Bench and the case for continuous benchmarks. *arXiv*, 2026.

- Nicolas Zucchet and **Scott W. Linderman**. Divergence controls entropy in distillation. *arXiv*, 2026.

- Aditi Jha, Keyue Sky Shi, Kara A. Fulton, Hanwen Lily Zhang, Christopher D. Harvey, Sandeep Robert Datta, Lisa M. Giocomo, and **Scott W. Linderman**. Inferring the evolving objectives that organize animal behavior. *bioRxiv*, 2026.

- Sandra A Romero Pinto, Jay A Hennig, Mark Burrell, Daigo Okada, Celia Benquet, Daniel Regester, Yoh Isogai, **Scott W Linderman**, Samuel J Gershman, and Naoshige Uchida. Dynamic belief representation and updating through learned attractor-like dynamics in the frontal cortex. *bioRxiv*, pages 2026–09, 2026.

- Henry D. Smith, Brian L. Trippe, and **Scott W. Linderman**. Closing the approximation gap in simulation-free latent SDEs. *arXiv 2606.16138*, 2026.

- Hyun Dong Lee, Aditi Jha, Stephen E. Clarke, Michael P. Silvernagel, Paul Nuyujukian, and **Scott W. Linderman**. Stiefel manifold dynamical systems for tracking representational drift. *bioRxiv*, 2026. *In press at Neural Computation.*

- Alisa D. Levin, Donald T. Avansino, Foram B. Kamdar, Nicholas S. Card, Maitreyee Wairagkar, Brandon G. Jacques, Justin J. Jude, Carrina Iacobacci, Bayardo E. Lacayo, Payton H. Bechefsky, Samuel R. Nason-Tomaszewski, Darrel R. Deo, Leigh R. Hochberg, Daniel B. Rubin, Ziv M. Williams, David M. Brandman, Sergey D. Stavisky, Nicholas AuYong, Chethan Pandarinath, **Scott W. Linderman**, Jaimie M. Henderson, and Francis R. Willett. Cross-brain transfer of high-performance intracortical speech and handwriting bcis. *bioRxiv*, 2026. *In revision to Nature Communications Biology.*

- Ari Beller, Yingchen Xu, Max H Siegel, Satchel Grant, Alan S Brown, Jan-Philipp Fränken, Joshua Tenenbaum, **Scott W Linderman**, and Tobias Gerstenberg. Multimodal inference through mental simulation, Sep 2025. *In submission to PNAS.*

- Shuyun Alina Xiao, Che Cherry Chen, Patricia Horvath, Valerie Tsai, Vibiana Marie Cardenas, Dan Biderman, Fei Deng, Yulong Li, **Scott W. Linderman**, Catherine Dulac, and Liqun Luo. Concerted actions of distinct serotonin neurons orchestrate female pup care behavior. *bioRxiv*, 2025. *Under review at Nature.*

- Christopher Versteeg, Jonathan D McCart, Mitchell Ostrow, David M Zoltowski, Clayton B Washington, Laura Driscoll, Olivier Codol, Jonathan A Michaels, **Scott W Linderman**, David Sussillo, and others. Computation-through-Dynamics benchmark: Simulated datasets and quality metrics for dynamical models of neural activity. *bioRxiv*, pages 2025–02, 2025. *Under review at PLoS CB.*

- Drew Friedmann, Xavier Gonzalez, Ashley Moses, Tanner Watts, Anthony Degleris, Nicole Ticea, Jun H Song, Sandeep Robert Datta, **Scott W Linderman***, and Liqun Luo*. Concerted modulation of spontaneous behavior and time-integrated whole-brain neuronal activity by serotonin receptors. *bioRxiv*, 2024.

- Rennie M Kendrick, **Scott Linderman**, and Scott F Owen. Transcriptomically-measured gene expression predicts physiological variation across single neurons in humans and mice. *bioRxiv*, pages 2024–08, 2024.

Peer reviewed journal, conference, and selected workshop papers

- Jun Ho Song, Drew Friedmann, Yunming Wu, Ashley Moses, Jalal Kenji Baruni, Yuan Yuan, Isabella Rana, Kelly Zhao, Tom Hindmarsh Sten, Qian Wang, Shuyun Alina Xiao, Xiaoke Chen, **Scott W. Linderman**, and Liqun Luo. Architecture of the serotonergic projectome in the mouse brain. *Cell*, 2026.

- Hyun Dong Lee, Xavier Gonzalez, Nicolas Zucchet, E. Kelly Buchanan, Emily Fox, and **Scott Linderman**. Switching linear attention. In *Third Conference on Language Modeling*, 2026.

- Aditya Nair, **Scott W. Linderman**, and David J. Anderson. The representation of affective internal states in the hypothalamus: A neural manifold view. *Annual Review Neuroscience*, 49:371–391, 2026.

- Amrith Lotlikar, Ian Christopher Tanoh, Praful K. Vasireddy, Andrew Lanpouthakoun, Michael A. Sommeling, Ramandeep Vilkhu, A.J. Phillips, Alexander Sher, Alan Litke, **Scott W. Linderman**, E. J. Chichilnisky, and Subhasish Mitra. Learning biophysical models of large-scale multineuronal data to enable precise neurostimulation. In *Forty-third International Conference on Machine Learning*, 2026.

- Xavier Gonzalez, E. Kelly Buchanan, Hyun Dong Lee, Jerry Weihong Liu, Ke Alexander Wang, David M. Zoltowski, Leo Kozachkov, Christopher Re, and **Scott Linderman**. A unifying framework for parallelizing sequential models with linear dynamical systems. *Transactions on Machine Learning Research*, 2026.

- Claire N Bedbrook, Ravi D Nath, Libby Zhang, **Scott W Linderman**, Anne Brunet, and Karl Deisseroth. Lifelong behavioral screen reveals an architecture of vertebrate aging. *Science*, 391(6790):eaea9795, 2026.

- **Randolph Linderman**, Noah Cowan, Yiran Chen, and **Scott Linderman**. A Bayesian nonparametric perspective on Mahalanobis distance for out of distribution detection. *Transactions on Machine Learning Research*, 2026.

- Caleb Weinreb, Lakshanyaa Thamarai Kannan, Alia Newman-Boulle, Tim Sainburg, Winthrop F. Gillis, Alex Plotnikoff, Sofia Makowska, Jonah E. Pearl, Mohammed Abdal Monium Osman, **Scott W. Linderman**, and Sandeep Robert Datta. Spontaneous behavior is a succession of self-directed tasks. *Neuron*, 2026.

- Yizi Zhang, Linyang He, Chaofei Fan, Tingkai Liu, Han Yu, Trung Le, Jingyuan Li, **Scott W Linderman**, Lea Duncker, Francis R Willett, Nima Mesgarani, and Liam Paninski. Decoding inner speech with an end-to-end brain-to-text neural interface. In *The Fourteenth International Conference on Learning Representations*, 2026.

- Shizhe He, Avanika Narayan, Ishan S Khare, **Scott W Linderman**, Christopher Ré, and Dan Biderman. An information theoretic perspective on agentic system design. In *The Fourteenth International Conference on Learning Representations*, 2026.

- Aditya Nair, Amit Vinograd, Mengyu Liu, George Mountoufaris, **Scott W Linderman**, and David J Anderson. The neural computation of affective internal states in the hypothalamus: A dynamical systems perspective. *Neuron*, 113(23):3887–3907, 2025.

- Xavier Gonzalez*, Leo Kozachkov*, David M. Zoltowski, Kenneth L. Clarkson, and **Scott W. Linderman**. Predictability enables parallelization of nonlinear state space models. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

- Amber Hu*, Henry Smith*, and **Scott W. Linderman**. SING: SDE inference via natural gradients. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

- David M. Zoltowski*, Skyler Wu*, Xavier Gonzalez, Leo Kozachkov, and **Scott W. Linderman**. Parallelizing MCMC across the sequence length. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

- Jon Saad-Falcon*, E. Kelly Buchanan*, Mayee F Chen*, Tzu-Heng Huang, Brendan McLaughlin, Tanvir Bhathal, Shang Zhu, Ben Athiwaratkun, Frederic Sala, **Scott W. Linderman**, Azalia Mirhoseini, and Christopher Re. Weaver: Shrinking the generation-verification gap by scaling compute for verification. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

- Ian Christopher Tanoh, Michael Deistler, Jakob H. Macke, and **Scott W. Linderman**. Identifying multi-compartment Hodgkin-Huxley models with high-density extracellular voltage recordings. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

- Yiqi Jiang, Kaiwen Sheng, Yujia Gao, E. Kelly Buchanan, Yu Shikano, Seung Je Woo, Yixiu Zhao, Tony Hyun Kim, Fatih Dinc, **Scott W. Linderman**, and Mark Schnitzer. Extracting task-relevant preserved dynamics from contrastive aligned neural recordings. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
  Selected for Spotlight Presentation. (3.2% of all submissions.)

- Yixiu Zhao, Jiaxin Shi, Feng Chen, Shaul Druckmann, Lester Mackey, and **Scott W. Linderman**. Informed correctors for discrete diffusion models. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

- Michael Bukwich, Malcolm G. Campbell, David Zoltowski, Lyle Kingsbury, Momchil S. Tomov, Joshua Stern, HyungGoo R. Kim, Jan Drugowitsch, **Scott W. Linderman**, and Naoshige Uchida. Competitive integration of time and reward explains value-sensitive foraging decisions and frontal cortex ramping dynamics. *Neuron*, 2025.

- Robin Vloeberghs, Anne E. Urai, Kobe Desender, and **Scott W. Linderman**. A Bayesian hierarchical model of trial-to-trial fluctuations in decision criterion. *PLoS Computational Biology*, 2025.

- Avanika Narayan*, Dan Biderman*, Sabri Eyuboglu*, Avner May, **Scott Linderman**, James Zou, and Christopher Re. Minions: Cost-efficient collaboration between on-device and cloud language models. *International Conference on Machine Learning (ICML)*, 2025.

- **Scott W. Linderman**, Peter Chang, Giles Harper-Donnelly, Aleyna Kara, Xinglong Li, Gerardo Duran-Martin, and Kevin Murphy. Dynamax: A Python package for probabilistic state space modeling with JAX. *Journal of Open Source Software*, 10(108):7069, April 2025.

- Francis R. Willett, Jingyuan Li, Trung Le, Chaofei Fan, Mingfei Chen, Eli Shlizerman, Yue Chen, Xin Zheng, Tatsuo S. Okubo, Tyler Benster, Hyun Dong Lee, Maxwell Kounga, E. Kelly Buchanan, David M. Zoltowski, **Scott W. Linderman**, and Jaimie M. Henderson. Brain-to-text benchmark '24: Lessons learned. *CoRR*, 2024.

- Amber Hu, David Zoltowski, Aditya Nair, David Anderson, Lea Duncker, and **Scott W. Linderman**. Modeling latent neural dynamics with Gaussian process switching linear dynamical systems. *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

- Julia C Costacurta, Shaunak Bhandarkar, David M Zoltowski, and **Scott W. Linderman**. Structured flexibility in recurrent neural networks via neuromodulation. *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

- Xavier Gonzalez, Andrew Warrington, Jimmy T. H. Smith, and **Scott W. Linderman**. Towards scalable and stable parallelization of nonlinear RNNs. *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

- George Mountoufaris, Aditya Nair, Bin Yang, Dong-Wook Kim, Amit Vinograd, Samuel Kim, **Scott W. Linderman**, and David J. Anderson. A line attractor encoding a persistent internal state requires neuropeptide signaling. *Cell*, 2024.

- Mengyu Liu*, Aditya Nair*, Nestor Coria, **Scott W. Linderman**, and David J. Anderson. Encoding of female mating dynamics by a hypothalamic line attractor. *Nature*, 2024.

- Amit Vinograd*, Aditya Nair*, Joseph Kim, **Scott W. Linderman**, and David J. Anderson. Causal evidence of a line attractor encoding an affective state. *Nature*, 2024.

- Samuel J Gershman, John A Assad, Sandeep Robert Datta, **Scott W Linderman**, Bernardo L Sabatini, Naoshige Uchida, and Linda Wilbrecht. Explaining dopamine through prediction errors and beyond. *Nature Neuroscience*, pages 1–11, 2024.

- Caleb Weinreb, Jonah E. Pearl, Sherry Lin, Mohammed Abdal Monium Osman, Libby Zhang, Sidharth Annapragada, Eli Conlin, Red Hoffmann, Sofia Makowska, Winthrop F. Gillis, Maya Jay, Shaokai Ye, Alexander Mathis, Mackenzie W. Mathis, Talmo Pereira, **Scott W. Linderman***, and Sandeep Robert Datta*. Keypoint-MoSeq: parsing behavior by linking point tracking to pose dynamics. *Nature Methods*, 21(7):1329–1339, 2024.

- Jakub Smékal, Jimmy T. H. Smith, Michael Kleinman, Dan Biderman, and **Scott W. Linderman**. Towards a theory of learning dynamics in deep state space models. In *ICML 2024 Workshop on Next Generation of Sequence Modeling Architectures*, 2024.
  Selected for Spotlight Presentation.

- Jimmy T.H. Smith, Shalini De Mello, Jan Kautz, **Scott W. Linderman**, and Wonmin Byeon. Convolutional state space models for long-range spatiotemporal modeling. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023.

- Jay Hennig, Sandra A. Romero Pinto, Takahiro Yamaguchi, **Scott W. Linderman**, Naoshige Uchida, and Samuel J. Gershman. Emergence of belief-like representations through reinforcement learning. *PLoS Computational Biology*, 2023.

- Yixin Wang, Anthony Degleris, Alex H Williams, and **Scott W Linderman**. Spatiotemporal clustering with Neyman-Scott processes via connections to Bayesian nonparametric mixture models. *Journal of the American Statistical Association*, 2023.

- Dieterich Lawson, Michael Y. Li, and **Scott W. Linderman**. NAS-X: Neural adaptive smoothing via twisting. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023.

- Hyun Dong Lee, Andrew Warrington, Joshua I Glaser, and **Scott W. Linderman**. Switching autoregressive low-rank tensor models. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023.

- Yixiu Zhao and **Scott W Linderman**. Revisiting structured variational autoencoders. *International Conference on Machine Learning (ICML)*, 2023.

- Jimmy TH Smith, Andrew Warrington, and **Scott W Linderman**. Simplified state space layers for sequence modeling. *International Conference on Learning Representations (ICLR)*, 2023.
  Selected for Oral Presentation. (1.5% of all submissions).

- Jeffrey Markowitz, Winthrop Gillis, Maya Jay, Jeffrey Wood, Ryley Harris, Robert Cieszkowski, Rebecca Scott, David Brann, Dorothy Koveal, Tomasz Kuila, Caleb Weinreb, Mohammed Osman, Sandra R Pinto, Naoschige Uchida, **Scott W Linderman**, Bernardo Sabatini, and Sandeep R Datta. Spontaneous behavior is structured by reinforcement without exogenous reward. *Nature*, 2023.

- Aditya Nair, Tomomi Karigo, Bin Yang, Surya Ganguli, Mark J Schnitzer, **Scott W Linderman**, David J Anderson, and Ann Kennedy. An approximate line attractor in the hypothalamus encodes an aggressive state. *Cell*, 186(1):178–193, 2023.

- Dieterich Lawson, Allan Raventos, Andrew Warrington, and **Scott W. Linderman**. SIXO: Smoothing inference with twisted objectives. *Advances in Neural Information Processing Systems*, 2022.
  Selected for Oral Presentation. (1.5% of all submissions).

- Julia C Costacurta, Lea Duncker, Blue Sheffer, Winthrop Gillis, Caleb Weinreb, Jeffrey Evan Markowitz, Sandeep R. Datta, Alex H Williams, and **Scott W. Linderman**. Distinguishing discrete and continuous behavioral variability using warped autoregressive HMMs. *Advances in Neural Information Processing Systems*, 2022.

- Ari Beller, Yingchen Xu, **Scott W Linderman**, and Tobias Gerstenberg. Looking into the past: Eye-tracking mental simulation in physical inference. In *Proceedings of the Annual Meeting of the Cognitive Science Society*, volume 44, 2022.

- Celia C. Beron, Shay Q. Neufeld, **Scott W. Linderman***, and Bernardo L. Sabatini*. Mice exhibit stochastic and efficient action switching during probabilistic decision making. *Proceedings of the National Academy of Sciences*, 119(15):e2113961119, 2022.

- Albert Lin, Daniel Witvliet, Luis Hernandez-Nunez, **Scott W Linderman**, Aravinthan D T Samuel, and Vivek Venkatachalam. Imaging whole-brain activity to understand behaviour. *Nature Reviews Physics*, pages 1–14, March 2022.

- **Scott W. Linderman**. Weighing the evidence in sharp-wave ripples. *Neuron*, 110(4):568–570, 2022.

- Alex H. Williams and **Scott W. Linderman**. Statistical neuroscience in the single trial limit. *Current Opinion in Neurobiology*, 70:193–205, 2021.

- Jimmy T. H. Smith, **Scott W. Linderman**, and David Sussillo. Reverse engineering recurrent neural networks with Jacobian switching linear dynamical systems. *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

- Alex H Williams, Erin Kunz, Simon Kornblith, and **Scott W. Linderman**. Generalized shape metrics on neural representations. *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

- Xinwei Yu, Matthew S Creamer, Francesco Randi, Anuj Kumar Sharma, **Scott W Linderman**, and Andrew Michael Leifer. Fast deep neural correspondence for tracking and identifying neurons in C. elegans using semi-synthetic training. *Elife*, 10:e66410, 2021.

- Isabel I C Low, Alex H Williams, Malcolm G Campbell, **Scott W Linderman**, and Lisa M Giocomo. Dynamic and reversible remapping of network representations in an unchanging environment. *Neuron*, August 2021.

- Libby Zhang, Jesse D Marshall, Timothy Dunn, Bence Ölveczky, and **Scott. W Linderman**. Animal pose estimation from video data with a hierarchical von Mises-Fisher-Gaussian model. *Proceedings of the International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2021.

- Arunesh Mittal, **Scott W. Linderman**, John Paisley, and Paul Sajda. Bayesian recurrent state space model for rs-fMRI. *Machine Learning for Health (ML4H) Workshop at NeurIPS 2020*, November 2020.

- Alex H Williams, Anthony Degleris, Yixin Wang, and **Scott W. Linderman**. Point process models for sequence detection in high-dimensional neural spike trains. *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
  Selected for Oral Presentation (1.1% of all submissions).

- Joshua I Glaser, Matthew Whiteway, John P Cunningham, Liam Paninski, and **Scott W. Linderman**. Recurrent switching dynamical systems models for multiple interacting neural populations. *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

- Wesley Tansey, Kathy Li, Haoran Zhang, **Scott W Linderman**, Raul Rabadan, David M Blei, and Chris H Wiggins. Dose-response modeling in high-throughput cancer drug screenings: An end-to-end approach. *Biostatistics*, 2020.

- David M. Zoltowski, Jonathan W. Pillow, and **Scott W. Linderman**. A general recurrent state space framework for modeling neural dynamics during decision-making. *Proceedings of the International Conference on Machine Learning (ICML)*, 2020.

- Robert Evan Johnson*, **Scott W Linderman***, Thomas Panier, Caroline Lei Wee, Erin Song, Kristian Joseph Herrera, Andrew Miller, and Florian Engert. Probabilistic models of larval zebrafish behavior reveal structure on many scales. *Current Biology*, 30:70–82, 2020.

- Ruoxi Sun*, **Scott W. Linderman***, Ian Kinsella, and Liam Paninski. Scalable Bayesian inference of dendritic voltage via spatiotemporal recurrent state space models. *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. **Selected for Oral Presentation (0.5% of all submissions).**

- Ifigeneia Apostolopoulou, **Scott W. Linderman**, Kyle Miller, and Artur Dubrawski. Mutually regressive point processes. *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

- Aaron Schein, **Scott W. Linderman**, Mingyuan Zhou, David Blei, and Hanna Wallach. Poisson-randomized gamma dynamical systems. *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

- Eleanor Batty*, Matthew Whiteway*, Shreya Saxena, Dan Biderman, Taiga Abe, Simon Musall, Winthrop Gillis, Jeffrey Markowitz, Anne Churchland, John Cunningham, **Scott W. Linderman**†, and Liam Paninski†. BehaveNet: nonlinear embedding and Bayesian neural decoding of behavioral videos. *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

- Josue Nassar, **Scott W. Linderman**, Monica Bugallo, and Il Memming Park. Tree-structured recurrent switching linear dynamical systems for multi-scale modeling. In *International Conference on Learning Representations (ICLR)*, 2019.

- Jeffrey E. Markowitz, Winthrop F. Gillis, Celia C. Beron, Shay Q. Neufeld, Keiramarie Robertson, Neha D. Bhagat, Ralph E. Peterson, Emalee Peterson, Minsuk Hyun, **Scott W. Linderman**, Bernardo L. Sabatini, and Sandeep Robert Datta. The striatum organizes 3D behavior via moment-to-moment action selection. *Cell*, May 2018.

- Anuj Sharma, Robert E. Johnson, Florian Engert, and **Scott W. Linderman**. Point process latent variable models of freely swimming larval zebrafish. *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

- Gonzalo E. Mena, David Belanger, **Scott W. Linderman**, and Jasper Snoek. Learning latent permutations with Gumbel-Sinkhorn networks. *International Conference on Learning Representations (ICLR)*, 2018.

- **Scott W. Linderman**, Gonzalo E. Mena, Hal Cooper, Liam Paninski, and John P. Cunningham. Reparameterizing the Birkhoff polytope for variational permutation inference. In *Proceedings of the 21st International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2018.

- Christian A. Naesseth, **Scott W. Linderman**, Rajesh Ranganath, and David M. Blei. Variational Sequential Monte Carlo. In *Proceedings of the 21st International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2018.

- **Scott W. Linderman***, Matthew J. Johnson*, Andrew C. Miller, Ryan P. Adams, David M. Blei, and Liam Paninski. Bayesian learning and inference in recurrent switching linear dynamical systems. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2017.

- Christian A. Naesseth, Francisco J. R. Ruiz, **Scott W. Linderman**, and David M. Blei. Reparameterization gradients through acceptance-rejection sampling algorithms. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2017.
  Received Best Paper Award.

- **Scott W. Linderman** and David M. Blei. Comment: A discussion of "Nonparametric Bayes modeling of populations of networks". *Journal of the American Statistical Association*, 112(520):1543–1547, 2017.

- **Scott W. Linderman** and Samuel J. Gershman. Using computational theory to constrain statistical models of neural data. *Current Opinion in Neurobiology*, 46:14 – 24, 2017.

- **Scott W. Linderman**, Ryan P. Adams, and Jonathan W. Pillow. Bayesian latent structure discovery from multi-neuron recordings. In *Advances in Neural Information Processing Systems (NIPS)*, 2016.

- Huseyin Melih Elibol, Vincent Nguyen, **Scott W. Linderman**, Matthew J. Johnson, Amna Hashmi, and Finale Doshi-Velez. Cross-corpora unsupervised learning of trajectories in autism spectrum disorders. *Journal of Machine Learning Research*, 17(133):1–38, 2016.

- **Scott W. Linderman**, Matthew J. Johnson, Matthew A. Wilson, and Zhe Chen. A Bayesian nonparametric approach to uncovering rat hippocampal population codes during spatial navigation. *Journal of Neuroscience Methods*, 263:36–47, 2016.

- **Scott W. Linderman***, Matthew J. Johnson*, and Ryan P. Adams. Dependent multinomial models made easy: Stick-breaking with the Pólya-gamma augmentation. In *Advances in Neural Information Processing Systems (NIPS)*, pages 3438–3446, 2015.

- **Scott W. Linderman**, Christopher H. Stock, and Ryan P. Adams. A framework for studying synaptic plasticity with neural spike train data. In *Advances in Neural Information Processing Systems (NIPS)*, pages 2330–2338, 2014.

- **Scott W. Linderman** and Ryan P. Adams. Discovering latent network structure in point process data. In *Proceedings of the International Conference on Machine Learning (ICML)*, pages 1413–1421, 2014.

Other preprints

- **Scott W. Linderman**, Annika L. A. Nichols, David M. Blei, Manuel Zimmer, and Liam Paninski. Hierarchical recurrent state space models reveal discrete and continuous dynamics of neural activity in C. elegans. *bioRxiv*, 2019.

- **Scott W. Linderman** and Ryan P. Adams. Scalable Bayesian inference for excitatory point process networks. *arXiv preprint arXiv:1507.03228*, 2015.

Student Advising

Current Students

- Cole Citrenbaum, PhD Student, Stanford Statistics 2025-
- Noah Cowan, PhD Student, Stanford Statistics 2024-
- Yiqi Jiang, PhD Student, Stanford Electrical Engineering 2025-
  Co-advised with Prof. Mark Schnitzer
- Hannah Lee, PhD Student, Stanford Electrical Engineering 2025-
- Hyun Dong Lee, PhD Student, Stanford Computer Science 2022-
  Co-advised with Prof. Emily Fox
- Alisa Levin, PhD Student, Stanford Computer Science 2022-
  Co-advised with Prof. Jaimie Henderson
- Jakub Smékal, PhD Student, Stanford Applied Physics 2024-
- Henry Smith, PhD Student, Stanford Statistics 2024-
  Co-advised with Prof. Brian Trippe
- Ian Christopher Tanoh, PhD Student, Stanford Statistics 2022-
- Skyler Wu, PhD Student, Stanford Statistics 2025-

Former Students

- Julia Costacurta, PhD Student, Stanford Electrical Engineering 2021-25
  Currently: Visiting Assistant Professor, Haverford College.
- Xavier Gonzalez, PhD Student, Stanford Statistics 2022-26
  Currently: Member of Technical Staff, Unconventional AI.
- Amber Hu, PhD Student, Stanford Statistics 2022-26
  Currently: Research Scientist, Datadog.
- Etaash Katiyar, PhD Student, Stanford Statistics 2024-26
- Dieterich Lawson, PhD Student, Stanford Computer Science 2020-23
  Currently: Research Scientist, Google.
- Matthew MacKay, PhD Student, Stanford Statistics 2021-23
  Currently: Technical Staff, Anthropic
- Blue Sheffer, PhD Student, Stanford Computer Science 2019-21
  Currently: Founder, School of Song.
- Jimmy Smith, PhD Student, Stanford ICME 2019-24
  Currently: Founding Scientist, Liquid AI
- Libby Zhang, PhD Student, Stanford Electrical Engineering 2019-25
  Currently: Shanahan Postdoctoral Fellow, Allen Institute for Brain Science.
- Yixiu Zhao, PhD Student, Stanford Applied Physics 2020-25
  Currently: Research Scientist, Basis Research Institute.

Postdoctoral Advising

Current Postdocs

- Kanha Batra, Postdoctoral Researcher, Stanford Neurobiology 2025-
  Co-advised with Prof. Lisa Giocomo
- Aditi Jha, Postdoctoral Researcher, Stanford Statistics 2024-
- Nicolas Zucchet, Postdoctoral Researcher, Stanford Statistics 2026-

Former Postdocs

- Dan Biderman, Postdoctoral Resarcher, Stanford Statistics and CS 2024-25
  Co-advised with Prof. Chris Ré.
  Currently: Co-Founder and CEO, Engram.

- E. Kelly Buchanan, Postdoctoral Researcher, Stanford Statistics 2024-26
  Co-advised with Prof. Chris Ré.
  Currently: Member of Technical Staff, NVIDIA.
- Lea Duncker, Postdoctoral Researcher, Stanford Statistics and EE 2021-23
  Co-advised with Prof. Krishna Shenoy.
  Currently: Assistant Professor, Columbia University.
- Elizabeth DuPre, Postdoctoral Researcher, Stanford Psychology 2022-24
  Co-advised with Prof. Russ Poldrack
  Currently: Postdoc, University of Montreal.
- Aditya Nair, Postdoctoral Researcher, Caltech 2024-25
  *Currently: Assistant Professor, LKCMedicine-NTU Singapore & IMCB-A*STAR.*
- Andrew Warrington, Postdoctoral Researcher, Stanford Statistics 2021-23
  Currently: Research Scientist at GE Healthcare
- Alex Williams, Postdoctoral Researcher, Stanford Statistics 2019-21
  Currently: Assistant Professor, NYU and Research Scientist, Flatiron Institute.
- David Zoltowski, Postdoctoral Researcher, Stanford Statistics 2022-26
  Co-advised with Dr. David Sussillo
  Currently: Research Scientist, Google DeepMind.

Research Support

Current Grants and Awards

- CAREER: State space models for neural and behavioral data
  Funding Agency: NSF
  Award Period: 08/01/2025 - 07/31/2030
  Contact PI: S. Linderman
  Role: PI

- The brain as a control center for other organs during aging
  Funding Agency: NIH
  Award Period: 12/01/2025 - 11/30/2030
  Contact PI: A. Brunet (Stanford)
  Role: MPI

- Understanding sensorimotor control through realistic neuro-biomechanical simulation
  Funding Agency: NIH
  Award Period: 04/01/2024 - 03/31/2027
  Contact PI: B. Olveczky (Harvard)
  Role: MPI

- Unified, Scalable, and Reproducible Neurostatistical Software
  Funding Agency: NIH
  Award Period: 07/01/23 - 06/30/2028
  Contact PI: A. Williams (NYU)
  Role: MPI

- Deconstructing the serotonin system in the mouse brain
  Funding Agency: NIH
  Award Period: 04/01/23 - 03/31/2028
  Contact PI: L. Luo
  Role: MPI

- CRCNS: Deonstructing the time-varying dynamics of motor cortex in freely moving behavior

Funding Agency: NSF/NIH
Award Period: 07/15/2022 - 04/30/2027
Contact PI: P. Nuyujukian (Stanford)
Role: Co-PI

- The dynamics of neural representations for distinct spatial contexts and memory episodes
  Funding Agency: NIH – NIMH
  Award Period: 04/01/2022 - 03/31/2027
  Contact PI: L. Giocomo (Stanford)
  Role: Co-I

Completed Grants and Awards

- Machine Learning Methods for Discovering Structure in Neural and Behavioral Data
  Funding Agency: McKnight Foundation
  Award Period: 07/01/23 - 06/30/2026
  Contact PI: S. Linderman
  Role: PI

- BRAID: DenPro3D Dendritic Processing of Spike Sequences in Biological and Artificial Brains
  Funding Agency: NSF
  Award Period: 07/01/2022 - 06/30/2026
  Contact PI: K. Boahen (Stanford)
  Role: Co-PI

- Probing the Dorsolateral Prefrontal Cortex and Central Executive Network for Improving Neuromodulation in Depression
  Funding Agency: CRCNS US-France Research Proposal
  Award Period: 07/1/2021 - 06/30/2026
  Contact PI: C. Keller (Stanford)
  Role: Co-I

- Discovering structure in neural and behavioral data
  Funding Agency: Alfred P. Sloan Foundation
  Award Period: 09/15/2022 - 09/14/2024
  Contact PI: S. Linderman
  Role: PI

- Toward a unified framework for dopamine signaling in the striatum
  Funding Agency: NIH NINDS (Natl. Inst. of Neurological Disorders and Stroke)
  Award Period: 8/15/2019 - 7/31/2024
  Contact PI: B. Sabatini (Harvard)
  Role: MPI

- Neural representation of mating partners by male C. elegans
  Funding Agency: NIH NINDS (Natl. Inst. of Neurological Disorders and Stroke)
  Award Period: 8/15/2019 - 7/31/2024
  Contact PI: P. Sternberg (Caltech)
  Role: MPI

- Dendritic Computation for Knowledge Systems
  Funding Agency: Stanford Institute for Human-Centered Artificial Intelligence
  Award Period: 09/01/2022 - 08/31/2023
  Contact PI: K. Boahen (Stanford)
  Role: Co-PI

- Discovering repeating neural motifs representing sequenced behavior
  Funding Agency: Simons Foundation (Simons Collaboration on the Global Brain)
  Award Period: 07/01/2020 - 06/30/2023
  Contact PI: B. Sabatini (Harvard)
  Role: Co-PI

- Multi-modal Inference in Brains, Minds, and Machines
  Funding Agency: Stanford Institute for Human-Centered Artificial Intelligence
  Award Period: 3/13/2019 - 3/12/2020
  Lead PI: T. Gerstenberg (Stanford)
  Role: Co-PI

- Scalable probabilistic inference for mechanistic models: Bridging the gap between scientific modelling and machine learning
  Funding Agency: Bavaria–California Technology Center (BaCaTeC)
  Award Period: 1/1/2019 - 6/1/2020
  Lead PI: J. Macke (Tubingen)
  Role: Co-PI

Teaching Experience

STATS305B: Applied Statistics II	2024-25
STATS305C: Applied Statistics III	2022,23,26
STATS320: ML Methods for Neural Data Analysis	2021,23,25
STATS271/371: Applied Bayesian Statistics	2021
STATS215: Statistical Models in Biology	2020

Professional Service

Summer School Organization and Teaching

Allen Institute Summer Workshop on the Dynamic Brain.	2021-
Woods Hole Methods in Computational Neuroscience Course.	2024-
Neuromatch Academy, Executive Committee (Co-Chair of Projects).	2022-23
JAX Short Course on Machine Learning for Quantifying Behavior.	2022-24

Conference and Workshop Organization

Co-Organizer, Statistical Analysis of Neural Data (SAND)	2025
Program Committee, Bernstein Conference	2024-25
Program Committee, COSYNE Conference	2022-24
Co-Organizer, COSYNE Workshop	2014,17,25
Co-Organizer, ICML Workshop	2020
Co-Organizer, NeurIPS Workshop	2019

Conference Area Chair

Neural Information Processing Systems (NeurIPS)	2019-
International Conference on Learning Representations (ICLR)	2021-23
International Conference on Machine Learning (ICML)	2021-23
Conference on Artificial Intelligence and Statistics (AISTATS)	2021-23

Journal and Conference Reviewing

- Annals of Applied Statistics
- Biometrika
- AAAI Conference on Artificial Intelligence (AAAI)
- eLife
- IEEE Transactions on Signal Processing
- International Conference on Machine Learning (ICML)
- International Conference on Artificial Intelligence and Statistics (AISTATS)

- Journal of the American Statistical Association (JASA)
- Journal of Computational Neuroscience
- Nature
- Nature Methods
- Nature Communications
- Neural Computation
- Neural Information Processing Systems (NeurIPS)
- PLoS Computational Biology
- SIAM Journal on Applied Dynamical Systems (SIADS)

Selected Talks

Neuroscience Theory Center, Columbia University	01/23/2026
Redwood Center for Theoretical Neuroscience, UC Berkeley	11/12/2025
Wharton Statistics Seminar, University of Pennsylvania	10/15/2025
Statistics Department, Duke University	09/19/2025
Neurobiology Department, Duke University	09/16/2025
Princeton Neuroscience Institute, Princeton University	10/24/2024
Jackson Laboratory	10/18/2024
Kempner Institute, Harvard University	10/11/2024
Chen Institute for Neuroscience Director's Seminar, Caltech	06/04/2024
University of Washington CoNectome Annual Symposium	05/13/2024
Two Sigma Investments	05/08/2024
Simons Collaboration on the Global Brain Annual Meeting	05/06/2024
International Winter Neuroscience Conference	04/09/2024
COSYNE Workshop on Reconstructing Dynamical Systems from Neural Data	03/04/2024
Statistics Seminar, Harvard University	11/27/2023
Neuroscience Seminar, Cold Spring Harbor Laboratory	11/20/2023
Biostatistics Seminar, University of North Carolina, Chapel Hill	09/07/2023
IDEAS Seminar Series in Theoretical Neuroscience, Georgia Tech	03/28/2023
COSYNE Workshop on Generative Models for Neural/Behavioral Data Analysis	03/14/2023
Machine Learning Seminar, Memorial Sloan-Kettering	12/06/2022
NeuroTheory Seminar, Columbia University	12/05/2022
Neuroscience Seminar, UCSD	09/27/2022
Neuroscience Seminar, University of Chicago	10/25/2022
Joint Statistics Meeting (JSM), Contributed Talk	08/01/2022
International Society for Bayesian Analysis, Contributed Talk	07/01/2022
Stanford-Berkeley Joint Statistics Colloquium, UC Berkeley	04/19/2022
Statistics Seminar, University of Connecticut	04/13/2022
External Seminar Series, Gatsby Unit, UCL	01/12/2022
Neuroscience Seminar, Boston University	11/03/2021
SymPOSEium, University of Minnesota	10/14/2021
CRCNS Workshop on Large Scale Neuroscience, NYU	10/09/2021
Bernstein Conference on Theoretical Neuroscience, Berlin	09/22/2021
Biostatistics Seminar, University of California, Berkeley	04/12/2021
Statistics Seminar, University of California, Los Angeles	03/04/2021
Statistics Seminar, University of British Columbia	11/03/2020
Biostatistics Seminar, Duke University	10/23/2020
MBCT Seminar, Stanford University	10/12/2020
ML Tea, Gatsby Computational Unit, University College London	04/24/2020
Statistics Seminar, University of California, Davis	04/23/2020
COSYNE Workshop on Interpretable Computational Neuroscience	03/01/2020

- Chan Zuckerberg Institute 01/31/2020
- Google Brain 01/22/2020
- Institute of Neuroscience Seminar. University of Oregon 11/21/2019
- Allen Institute for Brain Science. Seattle, WA 11/12/2019
- Biostatistics Workshop. Stanford University 10/24/2019
- COSYNE Workshop on Social Behavior. Lisbon, Portugal 03/05/2019
- COSYNE Workshop on Data, dynamics, and computation: Using data driven methods to ground mechanistic theory. 03/04/2019
- NeuroNex Workshop, Rice and Baylor University 01/31/2019
- Applied Math Seminar, Yale University 01/29/2019
- Artificial Intelligence Seminar, Cornell University 09/28/2018
- Center for Neuroengineering and Computation Seminar, Columbia University 09/24/2018
- Simons Collaboration on the Global Brain Postdoc Meeting 09/20/2018
- Air Force Research Laboratory, Rome, NY 08/27/2018
- IACS Seminar, Stony Brook University 07/13/2018
- Statistics Department, Stanford University 04/05/2018
- Allen Institute for Brain Science 03/28/2018
- Paul Allen School of Computer Science and Engineering, University of Washington 03/26/2018
- Janelia Farm Research Campus 02/28/2018
- Stanford Neurosciences Institute, Stanford University 02/20/2018
- Zuckerman Institute, Columbia University 02/13/2018
- Statistics Department, Columbia University 02/12/2018
- Biophysics Department, University of Washington 02/05/2018
- Center for Brain Science, Harvard University 01/30/2018